Exercise 1 - Dot-Product Attention

You are given a set of vectors

$$\mathbf{h}_1 = (1, 2, 3)^\top, \quad \mathbf{h}_2 = (1, 2, 1)^\top, \quad \mathbf{h}_3 = (0, 1, -1)^\top$$

and an alignment source vector $\mathbf{s} = (1, 2, 1)^\top$. Compute the resulting dot-product attention weights α_i for $i = 1, 2, 3$ and the resulting context vector $\mathbf{c}$.

Exercise 2 - Attention in Transformers

Transformers use a scaled dot product attention mechanism given by

$$C = \text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,$$

where $Q \in \mathbb{R}^{n_q \times d_k}$, $K \in \mathbb{R}^{n_k \times d_k}$, $V \in \mathbb{R}^{n_k \times d_v}$.

(a) Is the softmax function here applied row-wise or column-wise? What is the shape of the result?

(b) What is the value of d? Why is it needed?

(c) What is the computational complexity of this attention mechanism? How many additions and multiplications are required? Assume the canonical matrix multiplcation and not counting $\exp(x)$ towards computational cost.

(d) In the masked variant of the module, a masking matrix is added before the softmax function is applied. What are its values and its shape? For simplicity, assume $n_q = n_k$.

Exercise 3 - Scaled Dot-Product Attention by Hand

Consider the matrices Q, K, V given by

$$Q = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \quad K = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 2 & -2 \\ 1 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

Compute the context matrix C using the scaled dot product attention.